

XINING, China

WMO: 528660

Lat: 36.6644N

Lon: 101.7331E

Elev: 7906

StdP: 10.96

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.8	3.9	-12.8	3.7	21.2	-9.9	4.3	19.7	12.6	29.1	10.9	29.3	0.7	20	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	82.4	59.8	79.2	58.1	76.3	56.6	63.0	75.7	61.4	73.0	60.0	70.9	3.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.5	102.3	67.2	57.9	96.5	65.8	56.2	90.8	64.5	33.1	76.2	31.8	72.7	30.5	71.2	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	9.4	7.5	DB	-4.7	88.6	3.6	3.3	-7.3	91.0	-9.3	92.9	-11.3	94.7	-13.9	97.1	(4)
(5)			WB	-5.5	65.3	3.5	1.9	-8.0	66.7	-10.0	67.8	-12.0	68.9	-14.5	70.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	20.3	27.3	37.3	47.9	54.8	60.4	64.4	63.4	55.8	45.4	33.4	22.9	(6)
(7)		DBStd	16.15	5.45	6.32	6.60	6.47	5.16	4.11	4.68	4.79	4.58	5.02	6.11	5.02	(7)
(8)		HDD50	3580	920	636	396	109	17	0	0	0	6	157	499	840	(8)
(9)		HDD65	7574	1385	1056	859	512	317	146	67	90	279	609	949	1305	(9)
(10)		CDD50	1579	0	0	2	47	165	313	446	414	179	13	0	0	(10)
(11)		CDD65	98	0	0	0	0	0	8	49	39	2	0	0	0	(11)
(12)		CDH74	1188	0	0	1	17	54	158	562	365	30	1	0	0	(12)
(13)		CDH80	228	0	0	0	1	4	14	135	72	2	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.0	2.6	3.3	3.5	3.1	2.7	2.3	2.3	2.1	2.1	1.9	1.8	(14)
(15)	Precipitation	PrecAvg	16.7	0.1	0.1	0.4	0.9	2.2	2.7	3.3	3.4	2.6	0.9	0.2	0.0	(15)
(16)		PrecMax	21.3	0.2	0.3	1.1	2.7	4.3	4.4	5.9	6.4	4.6	2.3	0.6	0.2	(16)
(17)		PrecMin	10.8	0.0	0.0	0.0	0.2	0.9	1.5	1.8	1.5	0.6	0.3	0.0	0.0	(17)
(18)		PrecStd	2.2	0.0	0.1	0.3	0.6	0.9	0.8	0.8	0.9	0.9	0.4	0.1	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.3	55.8	67.7	76.7	79.0	82.1	89.1	86.5	78.0	69.4	59.0	47.7	(19)
(20)			MCWB	30.4	35.2	42.0	47.3	50.9	57.6	62.1	61.0	57.2	48.8	39.5	31.2	(20)
(21)		2%	DB	42.3	50.0	61.0	70.6	75.0	78.2	83.4	81.6	72.8	64.4	54.4	43.0	(21)
(22)			MCWB	28.0	32.5	39.0	45.6	50.5	55.6	60.4	60.4	55.2	47.0	37.1	29.0	(22)
(23)		5%	DB	38.1	45.5	56.2	66.4	71.6	75.3	80.0	78.0	69.4	60.4	50.7	39.7	(23)
(24)			MCWB	25.7	30.4	36.7	44.1	49.4	54.6	59.4	59.0	54.1	44.9	35.6	27.5	(24)
(25)		10%	DB	33.4	41.2	51.8	62.4	68.2	72.3	76.6	74.6	66.0	56.8	46.6	35.7	(25)
(26)			MCWB	23.2	28.4	35.2	42.4	48.4	53.8	58.6	58.1	52.7	43.7	33.8	25.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.6	36.0	43.1	50.3	54.8	61.6	65.4	65.1	61.4	51.3	40.7	31.8	(27)
(28)			MCD B	46.4	54.4	64.3	67.6	70.4	75.0	81.9	78.3	69.5	64.4	55.7	46.2	(28)
(29)		2%	WB	28.1	32.8	39.8	47.6	52.7	59.0	63.3	63.0	58.4	48.6	38.3	29.4	(29)
(30)			MCD B	42.0	48.6	57.2	65.5	68.1	71.3	77.5	74.6	68.2	59.4	51.8	42.2	(30)
(31)		5%	WB	25.8	30.8	37.8	45.6	51.2	57.4	61.6	61.6	56.1	46.7	36.4	27.6	(31)
(32)			MCD B	37.7	44.6	53.3	62.8	66.2	68.0	74.2	72.6	64.9	56.8	48.7	38.9	(32)
(33)		10%	WB	23.3	28.7	36.0	43.8	49.8	55.9	60.1	60.1	54.4	45.0	34.4	25.6	(33)
(34)			MCD B	33.1	40.5	50.0	59.5	64.7	66.8	71.7	70.6	63.1	54.5	45.7	35.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	28.0	27.5	25.8	24.3	23.0	22.1	21.3	20.8	19.5	22.8	25.6	27.0	(35)
(36)			MCD BR	35.7	35.6	35.4	32.5	30.7	28.9	29.3	27.9	26.8	30.2	32.3	33.1	(36)
(37)			MCWBR	23.3	21.0	17.2	13.1	10.8	9.9	9.5	9.1	10.5	13.6	17.0	20.8	(37)
(38)		5% WB	MCD BR	34.7	33.2	31.0	27.4	25.4	21.9	22.2	20.7	19.6	24.8	29.4	31.5	(38)
(39)			MCWBR	22.7	19.5	15.1	11.2	9.6	8.4	8.1	7.3	8.3	11.6	15.8	19.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.262	0.306	0.365	0.413	0.424	0.425	0.418	0.421	0.387	0.333	0.274	0.249	(40)	
(41)		taud	2.436	2.296	2.113	2.017	2.066	2.140	2.233	2.205	2.259	2.353	2.461	2.494	(41)	
(42)		Ebn at Noon	299	294	284	275	273	272	273	270	274	281	292	299	(42)	
(43)		Edh at Noon	27	35	46	54	52	49	44	44	40	33	26	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	988	1249	1570	1800	1859	1831	1772	1629	1315	1231	1045	917	(44)	
(45)		RadStd	58	53	87	92	104	95	163	112	103	100	62	34	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page